MILITARY SPECIFICATION

CATHETER AND NEEDLE UNIT, INTRAVENOUS, RADIOPAQUE, DISPOSABLE (CATHETER INTO NEEDLE)

This specification is approved for use within Defense Personnel Support Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers sterile, non-toxic, pyrogen-free, disposable, radiopaque catheter and needle intravenous placement units suitable for administering intravenous medication from a single venipuncture in pre-operative to post-operative treatment.

1.2 Classification. Catheter and needle units shall be furnished in the following classes, sizes, and styles as indicated below (see 6.1):

Class - Usable catheter length
- 1 - 12 inches (± 1 inch)
- 2 - 24 inches (± 1 inch)

Size - Catheter gage
- A - 16
- B - 18 or 18.5

Style - Needle gage
- a - 14
- b - 17

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-S-36586	- Sterilization Strip Test, Bacterial Spore.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Bureau of Biologics. Food and Drug Administration Code of Federal Regulation Part 273, Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockwell, MD 20852.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI MD70.1 American National Standard for Medical Material-Luer Taper Fittings - Performance.

(Application for copies should be addressed to Publication Sales, Dept. STD, American Institute of Physics, 335 East 54th St., NY, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F899 Standard Specification for Stainless Steel Billet, Bar, and Wire for Surgical Instruments.

ASTM F640 Standard Test Methods for Radiopacity of Plastics for Medical Use.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also maybe available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards). The text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. All plastic components to be in contact with patients's blood shall meet the requirements of the USP for Class IV plastic and transfusion and infusion assemblies. All plastic components shall have no more than 5 parts per million (ppm), cumulatively, extractable lead, barium, and calcium. The needle and wire stylet shall be fabricated from austenitic corrosion resistant steel in accordance with ASTM F899, and shall show no corrosion when subjected to the boil test specified in GG-I-526.

3.1.1 Catheter. The catheter shall be fabricated from clear, pliant polyvinylchloride or polytetrafluroethylene.

3.1.2 Needle. The needle shall be fabricated from seamless drawn, or welded and drawn, Type 304 corrosion-resistant steel.

3.1.3 Protective sleeve. The protective sleeve shall be fabricated from transparent or translucent flexible polyethylene.

3.1.4 Needle cover, needle collar, catheter hub and flow control plug. The needle cover, needle collar, catheter hub, and flow control plug shall be fabricated from rigid plastic.

3.2 Style and design. The catheter and needle intravenous unit shall be a sterile, nontoxic, pyrogen-free, disposable, radiopaque catheter placement unit, suitable for prolonged intravenous therapy from a single venipuncture. The unit shall be ready for use and shall permit aseptic insertion by the surgeon without assistance, and without the necessity of hand scrubbing or wearing of sterile gloves. The unit shall consist essentially of a 2 ± 1/8 inch long metal needle, of the short bevel design, an indwelling catheter that passes through the lumen of the needle, a catheter hub of Luer design, a wire stylet within the catheter, a flow control plug permanently attached to the wire stylet, needle collar, and a needle cover.

The catheter shall be permanently secured to a Luer-Lock hub, or alternatively, the catheter shall be one piece construction with the proximal end formed to a Luer-Lock hub. The distal edge of the catheter shall be smooth and free from burs and cuts. The needle hub shall make a secure fit when pushed into the needle collar. The needle collar shall be permanently affixed to the protective sleeve. The flow control plug shall be Luer tapered and provide a leak-proof seal with the catheter hub. The plug and stylet shall be removable from the catheter hub upon removal of the protective sleeve. Upon removal of the needle from the venipuncture site, the needle cover shall enclose ad lock the entire needle cannula, preventing the needle bevel from shearing the catheter.

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3.2.1 Catheter hub. The female Luer hub cavity of the catheter shall be tested for diameter and depth as specified in ANSI MD70.1. Hubs which do not fit within the limit planes of the male Luer test plug gage shall be rejected. In addition, when the hubs are fitted onto the plug gage there shall be no movement of the hubs on the gage when the hubs are pressed lightly from side to side in all directions.

3.2.2 Flow control plug. The male Luer taper of the flow control plug shall be tested for proper taper as specified in ANSI MD70.1. Male Luer tapers that do not fit within the limits of the Luer test gage shall be rejected.

3.3 Security of assembly.

3.3.1 Catheter hub to catheter. When tested as specified in 4.4.5.1, the catheter shall not break, loosen, or pull out of the catheter hub.

3.3.2 Flow control plug to wire stylet. When tested as specified in 4.4.5.2, the flow control plug shall not separate from the wire stylet.

3.3.3 Protective sleeve to needle collar. When tested as specified in 4.4.5.3, the needle collar shall maintain a continuous seal with the protective sleeve.

3.3.4 Needle to needle hub. When tested as specified in 4.4.7.5, the needle shall not separate from the needle hub.

3.4 Flexibility of catheter. When tested as specified in 4.4.6, the catheter shall not crack or collapse, nor shall liquid flow be obstructed in the lumen of the catheter.

3.5 X-ray opacity. When tested as specified in ASTM F640, the catheter shall appear prominent and conspicuous.

3.6 Leakage of catheter and catheter hub. When tested as specified in 4.4.7.1, the junction of the catheter and catheter hub shall show no leakage exceeding the formation of one clinging drop. As an alternate test, the above connection shall show no leakage exceeding the formation of one clinging air bubble when tested as specified in 4.4.7.2.

3.7 Freedom from pyrogens. All components of the unit shall be free from pyrogens when tested as specified in 4.4.4.

3.8 Sterility. All components of the unit shall be sterile when tested as specified in 4.4.3.

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3.8.1 Certificate of sterility. A certificate of sterility stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock Number (NSN).
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of test.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.8.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.9 Animal implantation test. The plastic catheter member only shall be tested, and shall show no evidence of toxicity or adverse tissue reaction upon gross and microscopic examination, when tested as specified in 4.4.2.1.

3.10 Instructions. A detailed instruction sheet for proper use of the catheter and needle units shall be provided in each immediate container. A caution note in the instructions shall state words to the effect:

"DO NOT ATTEMPT TO RETRACT CATHETER WHILE THE NEEDLE IS IN THE VEIN".

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3.11 Certificates of freedom from pyrogens, biological tests, and animal implantation test. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested in accordance with methods described in the USP and found to be free from pyrogens, and have passed the biological tests and the animal implantation test. This certificate shall include the specific test performed and the test results. The record of the animal implantation test and its results shall be certified by the pathologist who performed or supervised performance of the animal implantation test. These certificates shall be submitted by the contractor in the manner as specified in 3.8.1

3.12 Manufacturer's record. The manufacturer shall place a code or lot number on the immediate container which can be verified against the manufacturer's record of all test made on sterile, pyrogen-free units. These records shall be made available for inspection by any duly qualified official of the Federal Food and Drug Administration. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.13 Compatibility. The materials of which the catheter and needle units are manufactured shall not react with or be acted upon, by the liquids with which they will be in contact during normal use and operation nor shall there be any material incorporated in the units which will cause any deleterious effect when the units are used for the purpose intended.

3.14 Workmanship. Workmanship shall be first class throughout. The finished items shall be free from defects which detract from their appearance or impair their serviceability. The indwelling catheter, needle cannula and catheter hub shall show a smooth outer surface, free from pits, tool marks, burrs, scale, corrosion, and molding flash. Inside surfaces of all components of the unit shall be free from scale, deep pits, deep furrows and die marks. The lumen of the catheter, needle cannula, hub cavities, and protective conduit shall be free from burrs, chips, dirt, slivers, imbedded particles, or other foreign matter. Needles shall be sharp, not blunt or bent, and points shall be free from feathered edges, burrs or other defects, when examined under 10 power (10X) magnification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.4 Certificates of quality. Certificates of quality supplied by the manufacturer of polyvinylchloride, polytetrafluoroethylene, polyethylene, and corrosion-resistant steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacture or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table I. The unit of product for examination purposes shall be one complete catheter and needle unit.

TABLE I. Sampling criteria for examination.

Type defect	Inspection level	AQL (% defective)
Critical	III	1.0
Major	II	1.0

4.2.2 For Tests.

4.2.2.1 For sterility test. Sampling for sterility test shall be in accordance with the USP for parenteral devices and as specified in 4.4.5.

4.2.2.2 For pyrogen test. Sampling for pyrogen test shall be as indicated in the USP for transfusion and infusion assemblies. Samples shall be selected which are to be representative of each lot of plastic used in the fabrication of the units. As an alternate, sampling shall be on a sterilizer lot basis when sterilizer lots are smaller than, or equal to, production lots.

4.2.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. The unit of product for testing purposes shall be one complete catheter and needle unit. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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TABLE II. Sampling criteria for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (% defective)
	Composition analysis	3.1 & 3.1.2	ASTM F899	1/	--
	Boil test	3.1	GG-I-526	S-1	1.0
Needle Cannula	Dimensions	3.2	--	S-1	2.5
	Security of assembly	3.3.4	4.4.5.4	S-1	0.65
	Extractable metals	3.1	4.4.1	2	--
Catheter	Material	3.1.1	4.4.2	2/	--
	Dimensions	3.2	--	S-1	2.5
	Flexibility test	3.4	4.4.6	S-1	1.0
	X-ray opacity	3.5	ASTM F640	S-1	1.0
	Animal implantation test	3.9	4.4.2.1	--	--
	Leakage test	3.2	4.4.7	S-1	0.65
Catheter Hub	Extractable metals	3.1	4.4.1	2/	--
	Material	3.1.4	4.4.2	2/	--
	Diameter and depth of Luer cavity	3.2.1	ANSI MD70.1	S-1	0.65
	Security of assembly	3.3.1	4.4.5.1	S-1	0.65
Wire Stylet	Composition analysis	3.1	ASTM F899	1/	--
	Boil test	3.1	GG-I-526	S-1	1.0
Protective Sleeve	Security of assembly	3.3.3	4.4.5.3	S-1	0.65
Flow control Plug	Extractable metals	3.1	4.4.1	2/	--
	Material	3.1.4	4.4.2	2/	--
	Security of assembly	3.3.2	4.4.5.2	S-1	0.65
	Luer taper	3.2.2	ANSI MD70.1	S-1	0.65

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1/ Sufficient quantity of each lot of metal used in the construction of the wire stylet or needles shall be selected for chemical analysis.
2/ Sufficient samples to perform the extractable metal tests and biological tests shall be selected from each lot and shall be representative of each batch of plastic.

4.3 Examination.

4.3.1 Examination of catheter and needle units. The complete catheter and needle units shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any units in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Critical	
01	Catheter and catheter hub not permanently secured to each other.
02	Catheter pierced by needle.
03	Complete unit not free from foreign matter.
04	Protective sleeve and needle collar not permanently secured to each other.
Major	
101	Catheter and needle units not of the proper class, size, and style.
102	One or more components missing.
103	Catheter surface not free from kinks, dents, pits, blisters, cuts, holes, and scratches.
104	Needle point not keen, not straight, and not free from feathered edges.
105	Needle surfaces not free from dents, pits, burs, slivers, scale, and corrosion
106	Catheter does not form a safe, free sliding fit through the needle lumen.
107	Needle cannula not permanently secured to needle hub.
108	Needle cover does not completely encase the needle hub.
109	Protective sleeve opaque.
110	Wire stylet and flow control plug not permanently secure to each other.
111	Instructions for use and caution note missing.

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4.3.2 Packaging Inspection. The inspection of the unit package, packing, and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Extractable metals. The catheter, catheter hub, and flow control plug shall be tested for extractable metals. The specimens shall be cut such that no thickness exceeds 3 millimeters (mm) and eluted for either 72 hours at 50 degrees Centigrade (°C), 24 hours at 70°C, or 60 minutes at 121°C in two menstrua: 0.9 percent saline and cotton seed oil. Each gram of each specimen shall be covered with 5 milliliters (ml) of the menstruum (recommended amount is 10 grams of the specimens plus 50 ml of solvent). After agitation the menstrua shall be collected at the end of the selected elution period and submitted to spectrographic analysis for barium, cadmium, and lead. The metals shall be reported in parts per million (ppm) in the eluate. A value of 1.0 ppm in the eluate shall be equivalent to extracting 1.0 milligram from 200 grams of test material in the elution period into 1000 ml elution menstruum.

4.4.2 Biological tests for Class IV plastics. All plastic components coming in contact with intravenous medication and patient's blood shall be tested in accordance with the requirements of the USP for Class IV plastic and transfusion and infusion assemblies.

4.4.2.1 Animal implantation test. Samples of the finished indwelling plastic catheter (end item) shall be tested in accordance with the USP for Implantation Test, Plastic Containers, for Class IV plastics, except after "72 hours" add "and not to exceed 144 hours". Also delete "Examine macroscopically the area of the tissue surrounding the center portion of each implant strip. Use a magnifying lens if necessary" and substitute: "The muscle tissue surrounding the implants shall be examined, both grossly and microscopically, by a pathologist certified by the American Board of Pathology or the American College of Veterinary Pathologists."

4.4.3 Sterility Test

The catheter and needle unit shall be tested for sterility in accordance with the method specified in the U.S.P. using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium for the paper spore strip

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(carrier) shall be soybean-casein digest. The incubation media for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30° to 35° C for one half of the specimens and soybean-casein digest at 20° to 25° C for the remaining half of the specimens. The minimum incubation time shall be ten days. Sampling shall be as follows:

Medium	Paper Spore Strip	Inoc.Prod./Inoc.Simulated Prod
Soybean-casein digest	S-3*	S-3*
Fluid thioglycollate or Alternative thioglycollate		S-3*

*Use a minimum of 20 specimens. Acceptance number is zero.

4.4.3.1 Inoculated product or simulated product
For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism
Steam	B. stearothermophilus
Dry heat	B. stearothermophilus
	B. subtilis
Ethylene oxide	B. subtilis
Gamma radiation	B. pumilus

The number of viable spores shall be sufficient to provide assurance that the product has been sterilized.

4.4.3.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586A, Sterilization Test Strip Set, Bacterial Spore. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.3 Sterility test and 4.4.3.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.3.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P. and paragraphs 4.4.3 and 4.4.3.1.

4.4.3.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

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4.4.3.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the U.S.P. The inspection level shall be as stated in paragraph 4.4.3 above.

4.4.4 Freedom from pyrogens. Samples of the finished units shall be tested for freedom from pyrogens in accordance with the USP for transfusion and infusion assemblies.

4.4.5 Security of assembly tests.

4.4.5.1 Catheter hub to catheter. The security of assembly of the catheter hub to the catheter shall be tested by exerting a steady load of 5 pounds for a period of 5 minutes. The proximal end of the catheter shall be appropriately secured and the load shall be applied from the catheter hub, with the catheter hanging vertically.

4.4.5.2 Flow control plug to wire stylet. The security of assembly of the flow control plug to the wire stylet shall be tested by exerting a steady load of 3 pounds for a period of 5 minutes. The proximal end of the wire stylet shall be appropriately secured and the load shall be applied to the flow control plug, with the wire stylet and flow control plug hanging vertically.

4.4.5.3 Protective sleeve to needle collar. The security of assembly of the protective sleeve to the needle collar shall be tested by exerting a steady load of 3 pounds for a period of 3 minutes. The needle collar and protective sleeve shall hang vertically with the needle collar appropriately secured above its seal with the sleeve. The load shall be applied vertically from the distal end of the sleeve.

4.4.5.4 Needle to needle hub. The security of assembly of the needle cannula junction to the needle hub shall be tested by exerting a steady load of 10 pounds for a period of 5 minutes, with the needle hub appropriately secured.

4.4.6 Catheter flexibility test. The wire stylet shall be removed from the catheter. The plastic catheter shall be wrapped 360° around a circular rod or mandrel, 3/8 inch in diameter. The condition of the catheter shall be noted when released from the rod or mandrel. A suitable liquid shall then be allowed to flow thorough the lumen of the catheter within 5 seconds under an internal pressure of 3 ± 1/2 pounds per square inch gage (psig).

4.4.7 Leakage testing. The catheter shall be subjected to either the hydrostatic or pneumatic method for leakage testing.

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4.4.7.1 Leakage Test (hydrostatic method). The catheter hub shall be subjected to the water leakage test specified in ANSI Standard MD70.1. Prior to the test, water shall be admitted to flow through the lumen of the catheter so as to fill the catheter and catheter hub with the catheter subsequently sealed at its distal end in a leakproof manner.

4.4.7.2 Leakage test (pneumatic method). This test shall be performed similarly to the test specified in 4.4.8.1, except that air at 35 psig shall be the testing media. The complete assembly of the item to connected plug gage shall then be completely immersed into a suitable container of water, with the air pressure applied for at least 6 seconds. The entire unit shall be observed for leakage as evidence by air bubbles.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A

5.1.1.1 Individual package. Each catheter and needle unit shall be packaged in a sealed peel open type bag capable of maintaining sterility of contents unless the container is damaged or opened. The container shall be designed with tabs to facilitate opening. The unit shall be so packaged to prevent the tubing from being kinked or distorted.

5.1.1.2 Unit package. Twenty-five or fifty individual packages containing catheter and needle units of one size only shall be packaged in a snug fitting box constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for catheter and needles units shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Four unit packages containing catheter and needle units of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636 class weather-resistant. Closure, strapping and waterproofing shall be as specified in the applicable box specification.

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5.2.2 Level B. Four unit packages containing catheter and needle units of one size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for catheter and needle units shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Catheter and needle units packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. Catheter and needle units packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Individual package. In addition to any commercial markings, each individual package shall be marked with the lot (control) number, and the requirements of the Federal Food, Drug and Cosmetic Act. Marking shall include the following legend or similar statement: "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED"

5.3.2 Levels A, B and C.

5.3.2.1 Unit package. Each unit package shall be marked as specified in MIL-STD-129. Marking shall inlcude the lot (control) number, contract or purchase order number and name of contractor.

5.3.2.2 Exterior container and palletized load. Each exterior container and palletized load shall be marked as specified in MIL-STD-129. Marking shall include the lot (control) number.

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5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance wtih 4.3.2.

6. NOTES

6.1 Intended use. The catheter and needle unit shall be used for administering intravenous medication from a single venipuncture in pre-operative to post-operative treatment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type, class, size and style required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. Applicable National Stock Numbers.

6.3 This specification covers the following National Stock Numbers (NSNs):

NSN	Item Identification
6515-01-208-7328	CATHETER AND NEEDLE UNIT, INTRAVENOUS 16 Gage Catheter, Radiopaque, Disposable, 24 inches long, 25s
6515-01-208-7329	CATHETER AND NEEDLE UNIT, INTRAVENOUS, 18.5 Gage Catheter, Radiopaque, Disposable, 12 inches long, 50s

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NSN	Item Identification
6515-01-209-1174	CATHETER AND NEEDLE UNIT, INTRAVENOUS 16 Gage Catheter, Radiopaque, Disposable, 12 inches long, 50s
6515-01-209-1175	CATHETER AND NEEDLE UNIT, INTRAVENOUS 18.5 Gage Catheter, Radiopaque, Disposable, 24 inches long, 25s

MILITARY INTEREST

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:
DoD - MB

Agent
DLA-DM

Project No. 6515-3501

U.S. GOVERNMENT PRINTING OFFICE: 1986 - 605-036/41195

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-37994 (DLA-DM)	CATHETER AND NEEDLE UNIT, INTRAVENOUS, RADIOPAQUE, DISPOSABLE (CATHETER INTO NEEDLE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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